

SN74AVC1T45 Single-Bit Dual-Supply Bus Transceiver With Configurable Voltage Translation and 3-State Outputs

1 Features

- Available in the Texas Instruments NanoFree™ package
- Fully configurable dual-rail design allows each port to operate over the full 1.2V to 3.6V power-supply range
- V_{CC} isolation feature – if either V_{CC} input is at GND, then both ports are in the high-impedance state
- DIR input circuit referenced to V_{CCA}
- $\pm 12\text{mA}$ output drive at 3.3V
- I/Os are 4.6V tolerant
- I_{off} supports partial-power-down mode operation
- Typical maximum data rates
 - 500Mbps (1.8V to 3.3V translation)
 - 320Mbps (<1.8V to 3.3V translation)
 - 320Mbps (translate to 2.5V or 1.8V)
 - 280Mbps (translate to 1.5V)
 - 240Mbps (translate to 1.2V)
- Latch-up performance exceeds 100mA per JESD 78, Class II
- ESD protection exceeds JESD 22
 - $\pm 2000\text{V}$ Human Body Model (A114-A)
 - 200V Machine Model (A115-A)
 - $\pm 1000\text{V}$ Charged-Device Model (C101)

2 Applications

- [Personal electronic](#)
- [Industrial](#)
- [Enterprise](#)
- [Telecom](#)

3 Description

This single-bit noninverting bus transceiver uses two separate configurable power-supply rails. The SN74AVC1T45 is operational with V_{CCA}/V_{CCB} as low as 1.2V.

The A port is designed to track V_{CCA} . V_{CCA} accepts any supply voltage from 1.2V to 3.6V. The B port

is designed to track V_{CCB} . V_{CCB} accepts any supply voltage from 1.2V to 3.6V. This allows for universal low-voltage, bidirectional translation between any of the 1.2V, 1.5V, 1.8V, 2.5V, and 3.3V voltage nodes.

The SN74AVC1T45 is designed for asynchronous communication between two data buses. The logic levels of the direction-control (DIR) input activate either the B-port outputs or the A-port outputs. The device transmits data from the A bus to the B bus when the B-port outputs are activated and from the B bus to the A bus when the A-port outputs are activated. The input circuitry on both A and B ports always is active and must have a logic HIGH or LOW level applied to prevent excess I_{CC} and I_{CCZ} .

The SN74AVC1T45 is designed so that the DIR input is powered by V_{CCA} .

This device is fully specified for partial-power-down applications using I_{off} . The I_{off} circuitry disables the outputs, preventing damaging current backflow through the device when it is powered down.

The V_{CC} isolation feature is designed so that if either V_{CC} input is at GND, then both ports are in the high-impedance state.

NanoFree package technology is a major breakthrough in IC packaging concepts, using the die as the package.

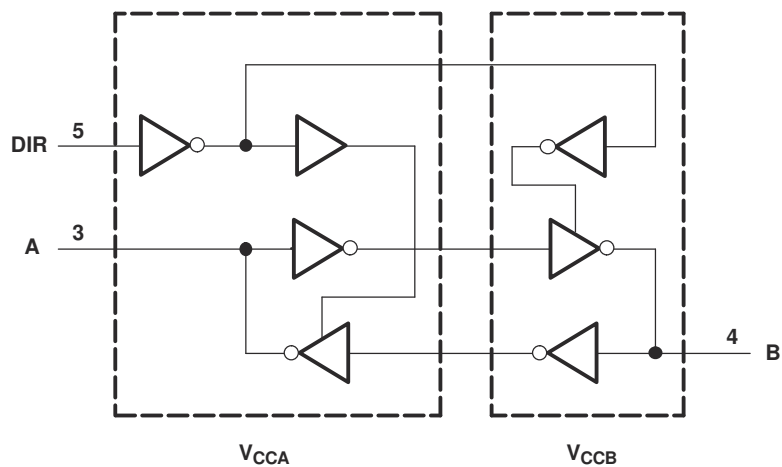
Package Information

PART NUMBER	PACKAGE ⁽¹⁾	PACKAGE SIZE ⁽²⁾
SN74AVC1T45	DCK (SOT, 6)	2mm × 2.1mm
	DBV (SOT-23, 6)	2.9mm × 2.8mm
	DRL (SOT-5X3, 6)	1.6mm × 1.6mm
	YZP (DSBGA, 6)	1.75mm × 1.25mm

(1) For more information, see [Section 11](#).

(2) The package size (length × width) is a nominal value and includes pins, where applicable.





Logic Diagram (Positive Logic)

Table of Contents

1 Features	1	7.1 Overview.....	15
2 Applications	1	7.2 Functional Block Diagram.....	15
3 Description	1	7.3 Feature Description.....	15
4 Pin Configuration and Functions	4	7.4 Device Functional Modes.....	15
5 Specifications	6	8 Application and Implementation	16
5.1 Absolute Maximum Ratings.....	6	8.1 Application Information.....	16
5.2 ESD Ratings.....	6	8.2 Typical Applications.....	16
5.3 Recommended Operating Conditions.....	7	8.3 Power Supply Recommendations.....	19
5.4 Thermal Information.....	8	8.4 Layout.....	19
5.5 Electrical Characteristics.....	8	9 Device and Documentation Support	21
5.6 Switching Characteristics, $V_{CCA} = 1.2V$	9	9.1 Receiving Notification of Documentation Updates....	21
5.7 Switching Characteristics, $V_{CCA} = 1.5 \pm 0.1V$	9	9.2 Support Resources.....	21
5.8 Switching Characteristics, $V_{CCA} = 1.8 \pm 0.15V$	10	9.3 Trademarks.....	21
5.9 Switching Characteristics, $V_{CCA} = 2.5 \pm 0.2V$	10	9.4 Electrostatic Discharge Caution.....	21
5.10 Switching Characteristics, $V_{CCA} = 3.3 \pm 0.3V$	11	9.5 Glossary.....	21
5.11 Operating Characteristics.....	11	10 Revision History	22
5.12 Typical Characteristics.....	12	11 Mechanical, Packaging, and Orderable Information	22
6 Parameter Measurement Information	14		
7 Detailed Description	15		

4 Pin Configuration and Functions

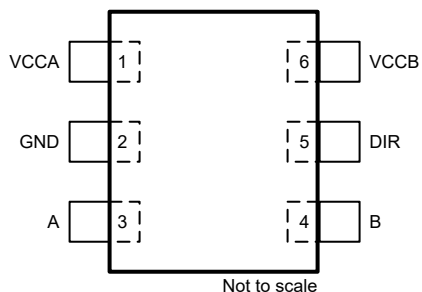


Figure 4-1. DBV Package, 6-Pin SOT-23 (Top View)

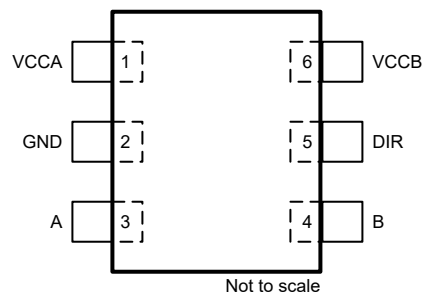


Figure 4-2. DCK Package, 6-Pin SOT-SC70 (Top View)

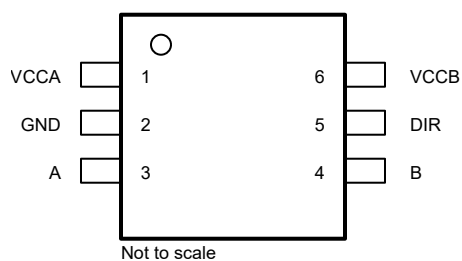


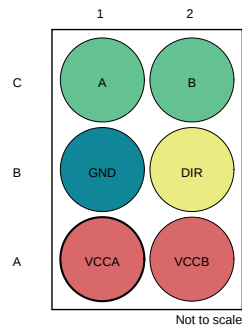
Figure 4-3. DRL Package, 6-Pin SOT-5X3 (Top View)

See mechanical drawings in [Section 11](#) for dimensions.

Table 4-1. Pin Functions

PIN		TYPE ⁽¹⁾	DESCRIPTION
NAME	NO.		
V _{CCA}	1	P	A-port supply voltage. $1.2V \leq V_{CCA} \leq 3.6V$
GND	2	G	Ground
A	3	I/O	Input/output A. Referenced to V _{CCA} .
B	4	I/O	Input/output B. Referenced to V _{CCB} .
DIR	5	I	Direction control signal
V _{CCB}	6	P	B-port supply voltage. $1.2V \leq V_{CCB} \leq 3.6V$.

(1) I = input, O = output, P = power, G = ground



**Figure 4-4. YZP Package,
6-Pin DSBGA (Bottom View)**

Legend	
Power	Input or Output
Ground	Input

Table 4-2. Pin Functions

PIN		TYPE ⁽¹⁾	DESCRIPTION
NO.	NAME		
A1	V _{CCA}	P	A-port supply voltage. 1.2V ≤ V _{CCA} ≤ 3.6V
A2	V _{CCB}	P	B-port supply voltage. 1.2V ≤ V _{CCB} ≤ 3.6V.
B1	GND	G	Ground
B2	DIR	I	Direction control signal
C1	A	I/O	Input/output A. Referenced to V _{CCA} .
C2	B	I/O	Input/output B. Referenced to V _{CCB} .

(1) I = input, O = output, P = power, G = ground

5 Specifications

5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted) ⁽¹⁾

			MIN	MAX	UNIT
V _{CCA}	Supply voltage A		−0.5	4.6	V
V _{CCB}	Supply voltage B		−0.5	4.6	V
V _I	Input Voltage ⁽²⁾	I/O Ports (A Port)	−0.5	4.6	V
		I/O Ports (B Port)	−0.5	4.6	
		Control Inputs	−0.5	4.6	
V _O	Voltage applied to any output in the high-impedance or power-off state ⁽²⁾	A Port	−0.5	4.6	V
		B Port	−0.5	4.6	
V _O	Voltage applied to any output in the high or low state ^{(2) (3)}	A Port	−0.5	V _{CCA} + 0.5	V
		B Port	−0.5	V _{CCB} + 0.5	
I _{IK}	Input clamp current	V _I < 0		−50	mA
I _{OK}	Output clamp current	V _O < 0		−50	mA
I _O	Continuous output current		−50	50	mA
	Continuous current through V _{CCA} , V _{CCB} , or GND		−100	100	mA
T _j	Junction Temperature			150	°C
T _{stg}	Storage temperature		−65	150	°C

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure beyond the limits listed in *Recommended Operating Conditions* may affect device reliability.
- (2) The input voltage and output negative-voltage ratings may be exceeded if the input and output current ratings are observed.
- (3) The output positive-voltage rating may be exceeded up to 6.5V maximum if the output current rating is observed.

5.2 ESD Ratings

			VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±2000	V
		Charged device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±1000	V
		Machine model, per A115-A	200	V

- (1) JEDEC document JEP155 states that 500V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250V CDM allows safe manufacturing with a standard ESD control process.

5.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted) ⁽¹⁾ ⁽²⁾ ⁽³⁾

			V _{CCI}	V _{CCO}	MIN	MAX	UNIT
V _{CCA}	Supply voltage A				1.2	3.6	V
V _{CCB}	Supply voltage B				1.2	3.6	
V _{IH}	High-level input voltage	Data inputs	1.2V to 1.95V		V _{CCI} × 0.65		V
			1.95V to 2.7V		1.6		V
			2.7V to 3.6V		2		V
V _{IL}	Low-level input voltage	Data inputs	1.2V to 1.95V		V _{CCI} × 0.35		V
			1.95V to 2.7V		0.7		V
			2.7V to 3.6V		0.8		V
V _{IH}	High-level input voltage	Control inputs (referenced to V _{CCA})	1.2V to 1.95V		V _{CCA} × 0.65		V
			1.95V to 2.7V		1.6		
			2.7V to 3.6V		2		
V _{IL}	Low-level input voltage	Control inputs (referenced to V _{CCA})	1.2V to 1.95V		V _{CCA} × 0.35		V
			1.95V to 2.7V		0.7		
			2.7V to 3.6V		0.8		
V _I	Input voltage ⁽³⁾				0	3.6	V
V _O	Output voltage	Active state			0	V _{CCO}	V
		3-state			0	3.6	
I _{OH}	High-level output current			1.2V		−3	mA
				1.4V to 1.6V		−6	
				1.65V to 1.95V		−8	
				2.3V to 2.7V		−9	
				3V to 3.6V		−12	
I _{OL}	Low-level output current			1.2V		3	mA
				1.4V to 1.6V		6	
				1.65V to 1.95V		8	
				2.3V to 2.7V		9	
				3V to 3.6V		12	
Δt/Δv	Input transition rise and fall time					5	ns/V
T _A	Operating free-air temperature				−40	85	°C

(1) V_{CCI} is the V_{CC} associated with the input port.

(2) V_{CCO} is the V_{CC} associated with the output port.

(3) All control inputs and data I/Os of this device have weak pulldowns to ensure the line is not floating when undefined external to the device. The input leakage from these weak pulldowns is defined by the I_I specification indicated under *Electrical Characteristics*.

5.4 Thermal Information

THERMAL METRIC ⁽¹⁾		SN74AVC1T45				UNIT
		DBV (SOT-23)	DCK (SOT-SC70)	DRL (SOT-5X3)	YZP (DSBGA)	
		6 PINS	6 PINS	6 PINS	6 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	183.4	211.4	236.2	130	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	102.5	145.2	135.2	54	
$R_{\theta JB}$	Junction-to-board thermal resistance	63.7	65.7	111.7	51	
Y_{JT}	Junction-to-top characterization parameter	39.5	47.0	16.5	1	
Y_{JB}	Junction-to-board characterization parameter	63.4	65.4	111	50	
$R_{\theta JC(bottom)}$	Junction-to-case (bottom) thermal resistance	N/A	N/A	N/A	N/A	

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application note.

5.5 Electrical Characteristics

over operating free-air temperature range (unless otherwise noted)^{(1) (2)}

PARAMETER		TEST CONDITIONS		V _{CCA}	V _{CCB}	Operating free-air temperature (T _A)						UNIT
						25°C			–40°C to 85°C			
						MIN	TYP	MAX	MIN	TYP	MAX	
V _{OH}		I _{OH} = –100μA	V _I = V _{IH}	1.2V to 3.6V	1.2V to 3.6V				V _{CCO} – 0.2			V
		I _{OH} = –3mA		1.2V	1.2V	0.95						
		I _{OH} = –6mA		1.4V	1.4V	1.05						
		I _{OH} = –8mA		1.65V	1.65V	1.2						
		I _{OH} = –9mA		2.3V	2.3V	1.75						
		I _{OH} = –12mA		3V	3V	2.3						
V _{OL}		I _{OL} = 100μA	V _I = V _{IL}	1.2V to 3.6V	1.2V to 3.6V				0.2			V
		I _{OL} = 3mA		1.2V	1.2V	0.15						
		I _{OL} = 6mA		1.4V	1.4V	0.35						
		I _{OL} = 8mA		1.65V	1.65V	0.45						
		I _{OL} = 9mA		2.3V	2.3V	0.55						
		I _{OL} = 12mA		3V	3V	0.7						
I _I	DIR	V _I = V _{CCA} or GND		1.2V to 3.6V	1.2V to 3.6V	–0.25	±0.025	0.25	–1	1	μA	
I _{off}	A port	V _I or V _O = 0V - 3.6V		0V	0V to 3.6V	–1	±0.1	1	–5	5	μA	
	B port			0V to 3.6V	0V	–1	±0.1	1	–5	5		
I _{OZ}	B port	V _I = or V _O = 0to 3.6V		0V	3.6V	–2.5	±0.5	2.5	–5	5	μA	
	A port			3.6V	0V	–2.5	±0.5	2.5	–5	5		
I _{CCA}		V _I = V _{CCI} or GND	I _O = 0	1.2V to 3.6V	1.2V to 3.6V				10			μA
				0V	3.6V				–2			
				3.6V	0V				10			
I _{CCB}		V _I = V _{CCI} or GND	I _O = 0	1.2V to 3.6V	1.2V to 3.6V				10			μA
				0V	3.6V				10			
				3.6V	0V				–2			
I _{CCA} + I _{CCB}		V _I = V _{CCI} or GND	I _O = 0	1.2V to 3.6V	1.2V to 3.6V				20			μA
C _i	Control Input	V _I = 3.3V or GND		3.3V	3.3V	2.5						pF

over operating free-air temperature range (unless otherwise noted)^{(1) (2)}

PARAMETER		TEST CONDITIONS		V _{CCA}	V _{CCB}	Operating free-air temperature (T _A)						UNIT
						25°C			–40°C to 85°C			
						MIN	TYP	MAX	MIN	TYP	MAX	
C _{io}	A or B port	V _O = 3.3V or GND		3.3V	3.3V	6						pF

- (1) V_{CCI} is the V_{CC} associated with the input port.
(2) V_{CCO} is the V_{CC} associated with the output port.

5.6 Switching Characteristics, $V_{CCA} = 1.2V$

over recommended operating free-air temperature range, $V_{CCA} = 1.2V$ (see Figure 6-1)

PARAMETER	FROM	TO	B-Port Supply Voltage (V_{CCB})					UNIT
			1.2V	1.5V	1.8V	2.5V	3.3V	
			TYP	TYP	TYP	TYP	TYP	
t_{PLH}	A	B	3.3	2.7	2.4	2.3	2.4	ns
t_{PHL}			3.3	2.7	2.4	2.3	2.4	
t_{PLH}	B	A	3.3	3.1	2.9	2.8	2.7	
t_{PHL}			3.3	3.1	2.9	2.8	2.7	
t_{PHZ}	DIR	A	5.1	5.2	5.3	5.2	3.7	ns
t_{PLZ}			5.1	5.2	5.3	5.2	3.7	
t_{PHZ}	DIR	B	5.3	4.3	4	3.3	3.7	
t_{PLZ}			5.3	4.3	4	3.3	3.7	
$t_{PZH}^{(1)}$	DIR	A	8.6	7.3	6.8	6.1	6.4	ns
$t_{PZL}^{(1)}$			8.6	7.3	6.8	6.1	6.4	
$t_{PZH}^{(1)}$	DIR	B	8.3	7.8	7.7	7.5	5.8	
$t_{PZL}^{(1)}$			8.3	7.8	7.7	7.5	5.8	

- (1) The enable time is a calculated value, derived using the formula shown in [Section 8.1.1](#).

5.7 Switching Characteristics, $V_{CCA} = 1.5 \pm 0.1V$

over recommended operating free-air temperature range, $V_{CCA} = 1.5 \pm 0.1V$ (see Figure 6-1)

PARAMETER	FROM	TO	B-Port Supply Voltage (V _{CCB})												UNIT															
			1.2V			1.5 ± 0.1V			1.8 ± 0.15V			2.5 ± 0.2V				3.3 ± 0.3V														
			MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX		MIN	TYP	MAX												
t _{PLH}	A	B	2.9			0.7			5.6			0.6			5.2			0.5			4.2			0.5			3.8			ns
t _{PHL}			2.9			0.7			5.6			0.6			5.2			0.5			4.2			0.5			3.8			
t _{PLH}	B	A	2.6			0.6			5.5			0.4			5.3			0.3			4.9			0.3			4.8			
t _{PHL}			2.6			0.6			5.5			0.4			5.3			0.3			4.9			0.3			4.8			
t _{PHZ}	DIR	A	3.8			1.6			6.7			1.5			6.8			0.3			6.9			0.9			6.9			ns
t _{PLZ}			3.8			1.6			6.7			1.5			6.8			0.3			6.9			0.9			6.9			
t _{PHZ}	DIR	B	5.1			1.8			8.1			1.6			7.1			1.1			4.7			1.4			4.5			
t _{PLZ}			5.1			1.8			8.1			1.6			7.1			1.1			4.7			1.4			4.5			
t _{PZH} ⁽¹⁾	DIR	A	7.7						13.6						12.4						9.6						9.3			ns
t _{PZL} ⁽¹⁾			7.7						13.6						12.4						9.6						9.3			
t _{PZH} ⁽¹⁾	DIR	B	6.7						12.3						12						11.1						10.7			
t _{PZL} ⁽¹⁾			6.7						12.3						12						11.1						10.7			

- (1) The enable time is a calculated value, derived using the formula shown in the [Section 8.1.1](#).

5.10 Switching Characteristics, $V_{CCA} = 3.3 \pm 0.3V$

over recommended operating free-air temperature range, $V_{CCA} = 3.3 \pm 0.3V$ (see Figure 6-1)

PARAMETER	FROM	TO	B-Port Supply Voltage (V _{CCB})												UNIT															
			1.2V			1.5 ± 0.1V			1.8 ± 0.15V			2.5 ± 0.2V				3.3 ± 0.3V														
			MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX		MIN	TYP	MAX												
t _{PLH}	A	B	2.6			0.4			4.7			0.3			4.4			0.2			3.3			0.2			2.8			ns
t _{PHL}			2.6			0.4			4.7			0.3			4.4			0.2			3.3			0.2			2.8			
t _{PLH}	B	A	2.2			0.4			3.8			0.3			3.4			0.2			3			0.1			2.8			
t _{PHL}			2.2			0.4			3.8			0.3			3.4			0.2			3			0.1			2.8			
t _{PHZ}	DIR	A	3.1			1.3			4.3			1.3			4.3			1.3			4.3			1.3			4.3			ns
t _{PLZ}			3.1			1.3			4.3			1.3			4.3			1.3			4.3			1.3			4.3			
t _{PHZ}	DIR	B	4			0.7			7.4			0.6			6.5			0.7			4			1.5			4.9			
t _{PLZ}			4			0.7			7.4			0.6			6.5			0.7			4			1.5			4.9			
t _{PZH} ⁽¹⁾	DIR	A	6.2						11.2						9.9						7						6.7			ns
t _{PZL} ⁽¹⁾			6.2						11.2						9.9						7						6.7			
t _{PZH} ⁽¹⁾	DIR	B	5.7						8.9						8.5						7.2						6.8			
t _{PZL} ⁽¹⁾			5.7						8.9						8.5						7.2						6.8			

(1) The enable time is a calculated value, derived using the formula shown in the [Section 8.1.1](#).

5.11 Operating Characteristics

$T_A = 25^\circ C$

PARAMETER		TEST CONDITIONS	B-Port Supply Voltage (V_{CCB})					UNIT
			1.2V	1.5V	1.8V	2.5V	3.3V	
			TYP	TYP	TYP	TYP	TYP	
$C_{pdA}^{(1)}$	A-port input, B-port output	$C_L = 0pF$, $f = 10MHz$, $t_r = t_f = 1ns$	3	3	3	3	4	pF
	A-port input, B-port output		13	13	14	15	15	
$C_{pdB}^{(1)}$	A-port input, B-port output	$C_L = 0pF$, $f = 10MHz$, $t_r = t_f = 1ns$	13	13	14	15	15	pF
	A-port input, B-port output		3	3	3	3	3	

(1) Power dissipation capacitance per transceiver.

5.12 Typical Characteristics

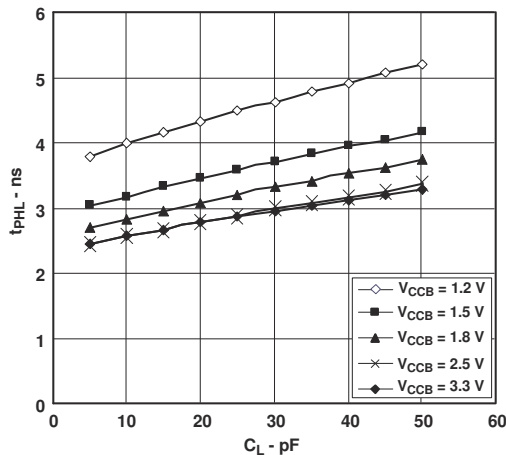


Figure 5-1. Typical Propagation Delay of High-to-Low (A to B) vs Load Capacitance
 $T_A = 25^\circ\text{C}$, $V_{CCA} = 1.2\text{V}$

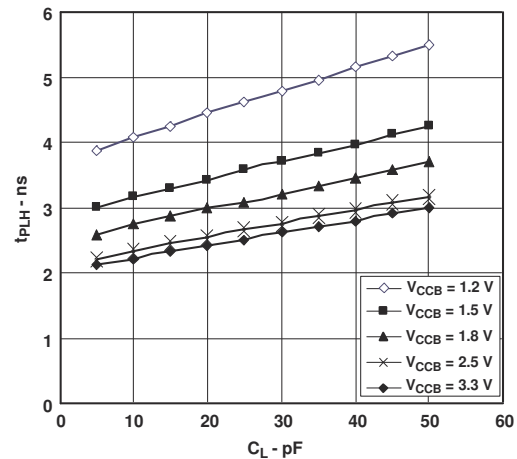


Figure 5-2. Typical Propagation Delay of Low-to-High (A to B) vs Load Capacitance
 $T_A = 25^\circ\text{C}$, $V_{CCA} = 1.2\text{V}$

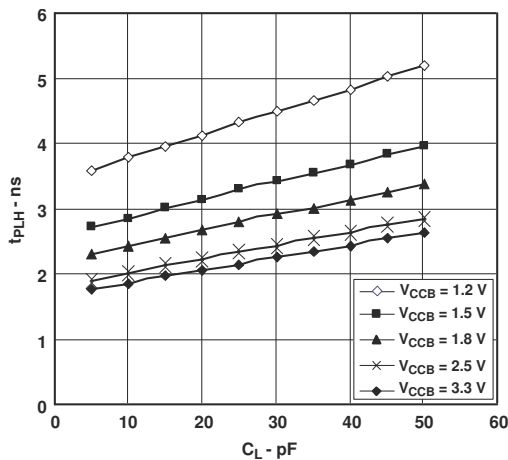


Figure 5-3. Typical Propagation Delay of High-to-Low (A to B) vs Load Capacitance
 $T_A = 25^\circ\text{C}$, $V_{CCA} = 1.5\text{V}$

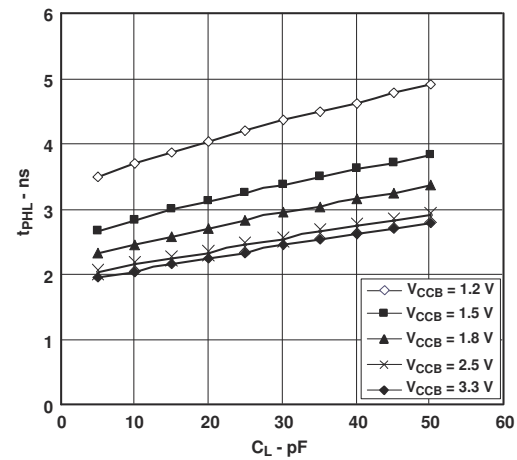


Figure 5-4. Typical Propagation Delay of Low-to-High (A to B) vs Load Capacitance
 $T_A = 25^\circ\text{C}$, $V_{CCA} = 1.5\text{V}$

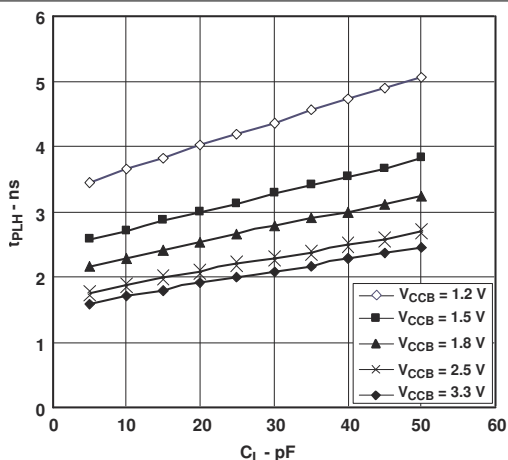


Figure 5-5. Typical Propagation Delay of High-to-Low (A to B) vs Load Capacitance
 $T_A = 25^\circ\text{C}$, $V_{CCA} = 1.8\text{V}$

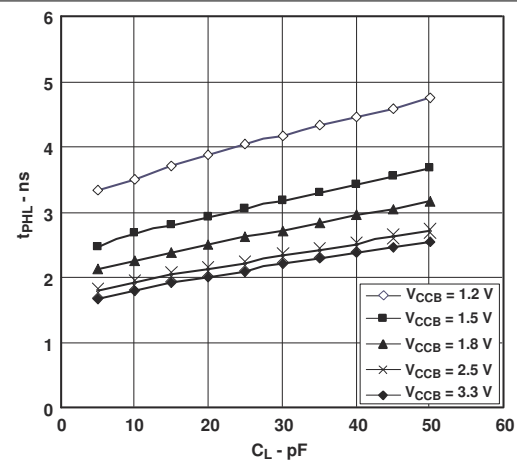


Figure 5-6. Typical Propagation Delay of Low-to-High (A to B) vs Load Capacitance
 $T_A = 25^\circ\text{C}$, $V_{CCA} = 1.8\text{V}$

5.12 Typical Characteristics (continued)

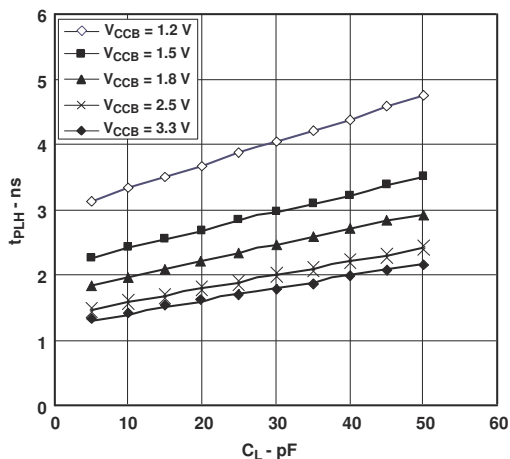


Figure 5-7. Typical Propagation Delay of High-to-Low (A to B) vs Load Capacitance
 $T_A = 25^\circ\text{C}$, $V_{CCA} = 2.5\text{V}$

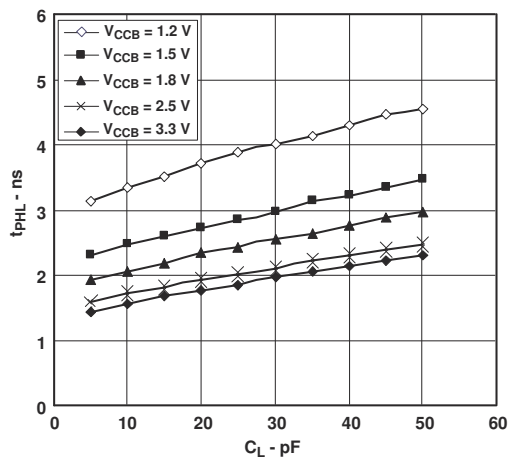


Figure 5-8. Typical Propagation Delay of Low-to-High (A to B) vs Load Capacitance
 $T_A = 25^\circ\text{C}$, $V_{CCA} = 2.5\text{V}$

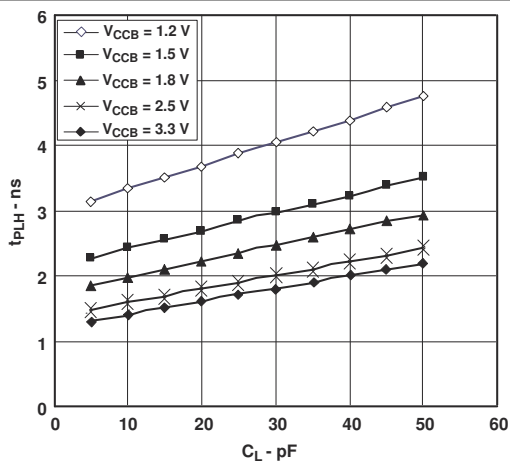


Figure 5-9. Typical Propagation Delay of High-to-Low (A to B) vs Load Capacitance
 $T_A = 25^\circ\text{C}$, $V_{CCA} = 3.3\text{V}$

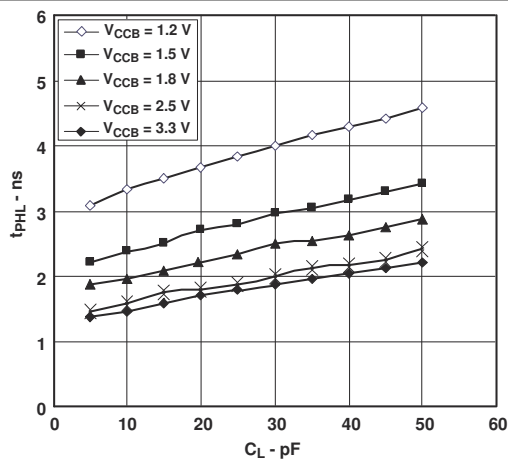
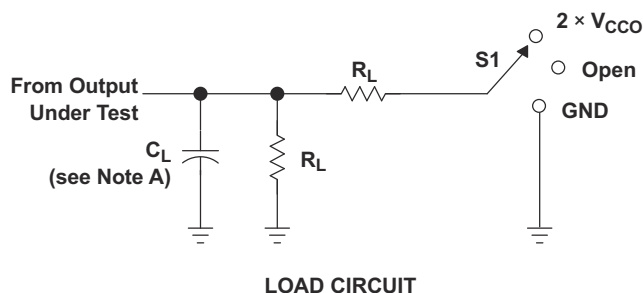


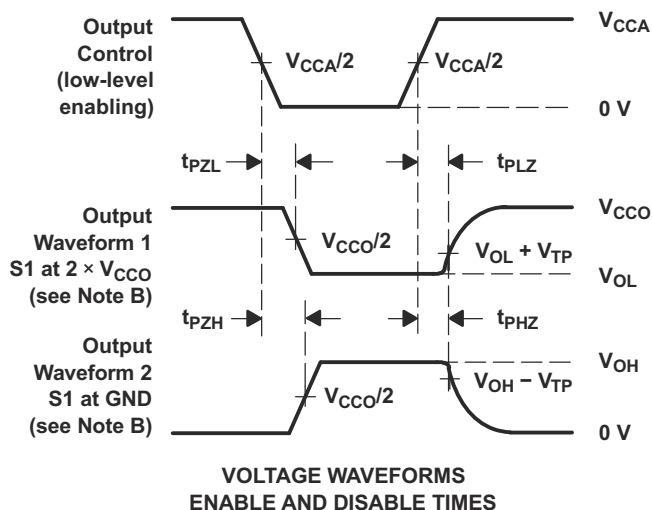
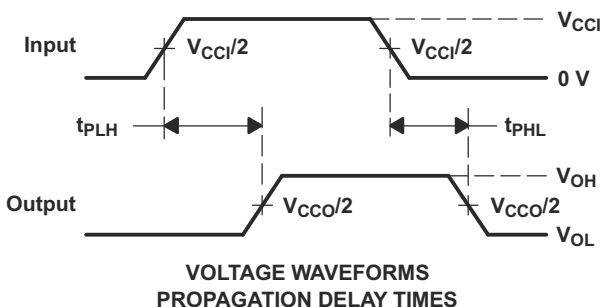
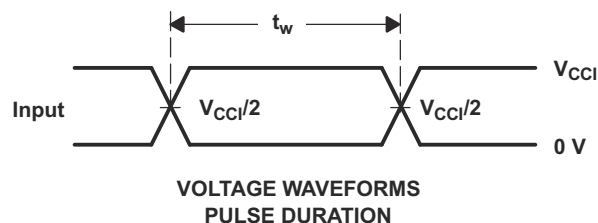
Figure 5-10. Typical Propagation Delay of Low-to-High (A to B) vs Load Capacitance
 $T_A = 25^\circ\text{C}$, $V_{CCA} = 3.3\text{V}$

6 Parameter Measurement Information



V_{CCO}	C_L	R_L	V_{TP}
1.2 V	15 pF	2 kW	0.1 V
1.5 V ± 0.1 V	15 pF	2 kW	0.1 V
1.8 V ± 0.15 V	15 pF	2 kW	0.15 V
2.5 V ± 0.2 V	15 pF	2 kW	0.15 V
3.3 V ± 0.3 V	15 pF	2 kW	0.3 V

TEST	S1
t_{pd}	Open
t_{PLZ}/t_{PZL}	$2 \times V_{CCO}$
t_{PHZ}/t_{PZH}	GND



- NOTES: A. C_L includes probe and jig capacitance.
- B. Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high, except when disabled by the output control.
- C. All input pulses are supplied by generators having the following characteristics: PRR = 10 MHz, $Z_O = 50 \Omega$, $dv/dt \geq 1 \text{ V/ns}$.
- D. The outputs are measured one at a time, with one transition per measurement.
- E. t_{PLZ} and t_{PHZ} are the same as t_{dis} .
- F. t_{PZL} and t_{PZH} are the same as t_{en} .
- G. t_{PLH} and t_{PHL} are the same as t_{pd} .
- H. V_{CCI} is the V_{CC} associated with the input port.
- I. V_{CCO} is the V_{CC} associated with the output port.

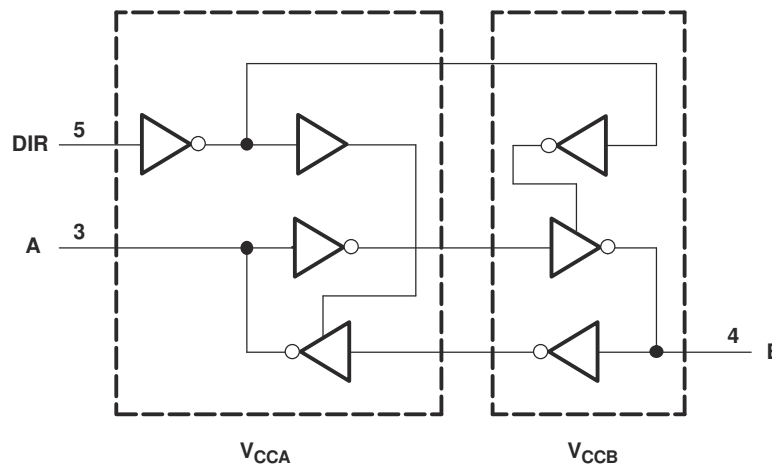
Figure 6-1. Load Circuit and Voltage Waveforms

7 Detailed Description

7.1 Overview

The SN74AVC1T45 is a single-bit, dual-supply, noninverting voltage level translation device. V_{CCA} supports pin A and the direction control pin, and V_{CCB} supports pin B. The A port can accept I/O voltages ranging from 1.2V to 3.6V, while the B port can accept I/O voltages from 1.2V to 3.6V. The high on DIR allows data transmission from A to B and a low on DIR allows data transmission from B to A.

7.2 Functional Block Diagram



7.3 Feature Description

7.3.1 Fully Configurable

The fully configurable dual-rail design allows each port to operate over the full 1.2V to 3.6V power-supply range. Both V_{CCA} and V_{CCB} can be supplied at any voltage between 1.2V and 3.6V making the device an excellent choice for translating between any of the voltage nodes (1.2V, 1.8V, 2.5V, and 3.3V).

7.3.2 Support High-Speed Translation

SN74AVC1T45 can support high data-rate application. The translated signal data rate can be up to 500Mbps when signal is translated from 1.8V to 3.3V.

7.3.3 I_{off} Supports Partial-Power-Down Mode Operation

I_{off} will prevent backflow current by disabling I/O output circuits when device is in partial-power-down mode.

7.4 Device Functional Modes

Table 7-1. Function Table

INPUT DIR ⁽¹⁾	OPERATION
L	B data to A bus
H	A data to B bus

(1) Input circuits of the data I/Os always are active.

8 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

8.1 Application Information

The SN74AVC1T45 device can be used in level-translation applications for interfacing devices or systems operating at different interface voltages with one another. The maximum data rate can be up to 500Mbps when device translate signal from 1.8V to 3.3V.

8.1.1 Enable Times

Calculate the enable times for the SN74AVC1T45 using the following formulas:

- $t_{PZH} \text{ (DIR to A)} = t_{PLZ} \text{ (DIR to B)} + t_{PLH} \text{ (B to A)}$
- $t_{PZL} \text{ (DIR to A)} = t_{PHZ} \text{ (DIR to B)} + t_{PHL} \text{ (B to A)}$
- $t_{PZH} \text{ (DIR to B)} = t_{PLZ} \text{ (DIR to A)} + t_{PLH} \text{ (A to B)}$
- $t_{PZL} \text{ (DIR to B)} = t_{PHZ} \text{ (DIR to A)} + t_{PHL} \text{ (A to B)}$

In a bidirectional application, these enable times provide the maximum delay from the time the DIR bit is switched until an output is expected. For example, if the SN74AVC1T45 initially is transmitting from A to B, then the DIR bit is switched; the B port of the device must be disabled before presenting it with an input. After the B port has been disabled, an input signal applied to it appears on the corresponding A port after the specified propagation delay.

8.2 Typical Applications

8.2.1 Unidirectional Logic Level-Shifting Application

Figure 8-1 shows an example of the SN74AVC1T45 being used in a unidirectional logic level-shifting application.

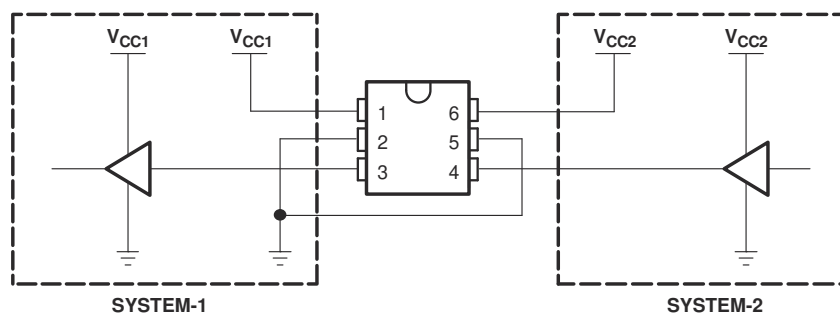


Figure 8-1. Unidirectional Logic Level-Shifting Application

PIN	NAME	FUNCTION	DESCRIPTION
1	V _{CCA}	V _{CC1}	SYSTEM-1 supply voltage (1.2V to 3.6V)
2	GND	GND	Device GND
3	A	OUT	Output level depends on V _{CC1} voltage.
4	B	IN	Input threshold value depends on V _{CC2} voltage.
5	DIR	DIR	GND (low level) determines B-port to A-port direction.
6	V _{CCB}	V _{CC2}	SYSTEM-2 supply voltage (1.2V to 3.6V)

8.2.1.1 Design Requirements

For this design example, use the parameters listed in [Table 8-1](#).

Table 8-1. Design Parameters

DESIGN PARAMETERS	EXAMPLE VALUES
Input voltage range	1.2V to 3.6V
Output voltage range	1.2V to 3.6V

8.2.1.2 Detailed Design Procedure

To begin the design process, determine the following:

- Input voltage range
 - Use the supply voltage of the device that is driving the SN74AVC1T45 device to determine the input voltage range. For a valid logic-high, the value must exceed the V_{IH} of the input port. For a valid logic low the value must be less than the V_{IL} of the input port.
- Output voltage range
 - Use the supply voltage of the device that the SN74AVC1T45 device is driving to determine the output voltage range.

8.2.1.3 Application Curve

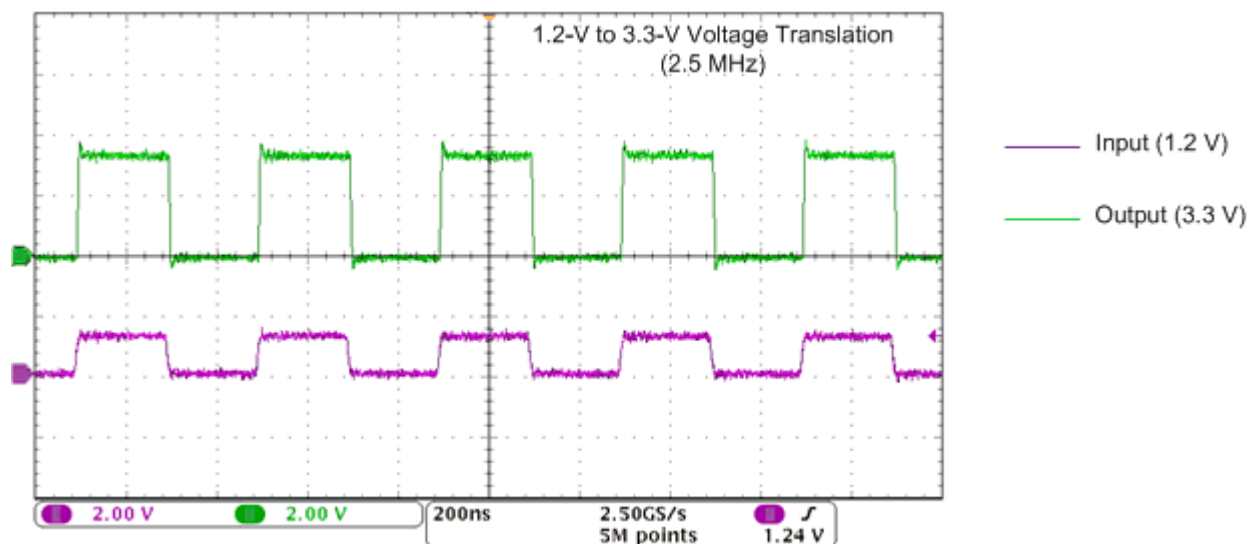


Figure 8-2. Translation Up (1.2V to 3.3V) at 2.5MHz

8.2.2 Bidirectional Logic Level-Shifting Application

Figure 8-3 shows the SN74AVC1T45 being used in a bidirectional logic level-shifting application. Because the SN74AVC1T45 does not have an output-enable (OE) pin, the system designer should take precautions to avoid bus contention between SYSTEM-1 and SYSTEM-2 when changing directions.

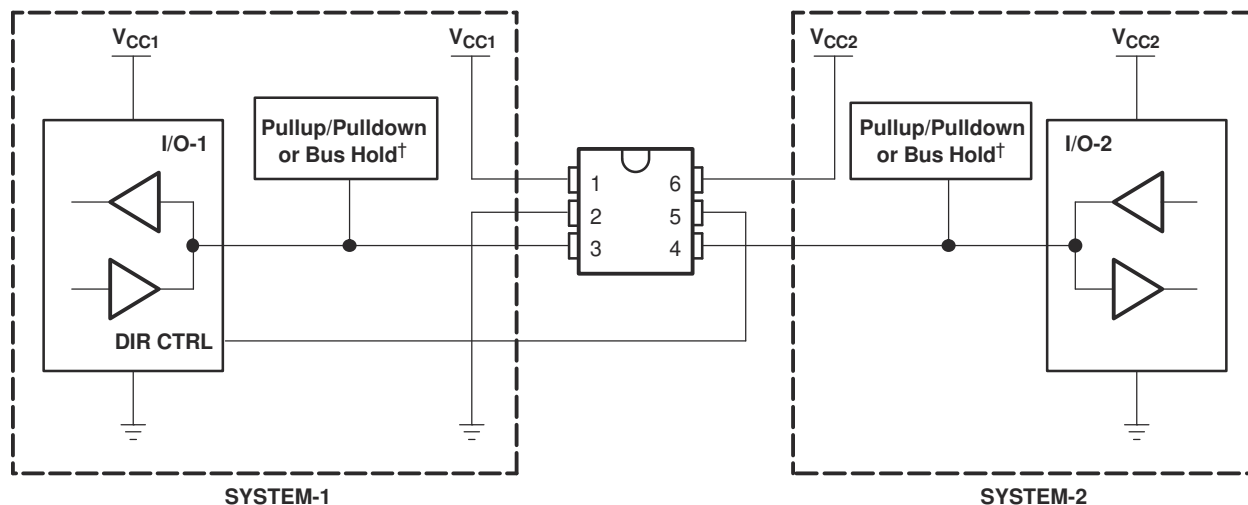


Figure 8-3. Bidirectional Logic Level-Shifting Application

The following table provides data transmission from SYSTEM-1 to SYSTEM-2 and then from SYSTEM-2 to SYSTEM-1.

Table 8-2. Data Transmission: SYSTEM-1 and SYSTEM-2

STATE	DIR CTRL	I/O-1	I/O-2	DESCRIPTION
1	H	Out	In	SYSTEM-1 data to SYSTEM-2
2	H	Hi-Z	Hi-Z	SYSTEM-2 is getting ready to send data to SYSTEM-1. I/O-1 and I/O-2 are disabled. The bus-line state depends on pullup or pulldown. ⁽¹⁾
3	L	Hi-Z	Hi-Z	DIR bit is flipped. I/O-1 and I/O-2 still are disabled. The bus-line state depends on pullup or pulldown. ⁽¹⁾
4	L	In	Out	SYSTEM-2 data to SYSTEM-1

(1) SYSTEM-1 and SYSTEM-2 must use the same conditions (for example, both pullup or both pulldown).

8.2.2.1 Design Requirements

Refer to [Section 8.2.1.1](#).

8.2.2.2 Detailed Design Procedure

Refer to [Section 8.2.1.2](#).

8.2.2.3 Application Curve

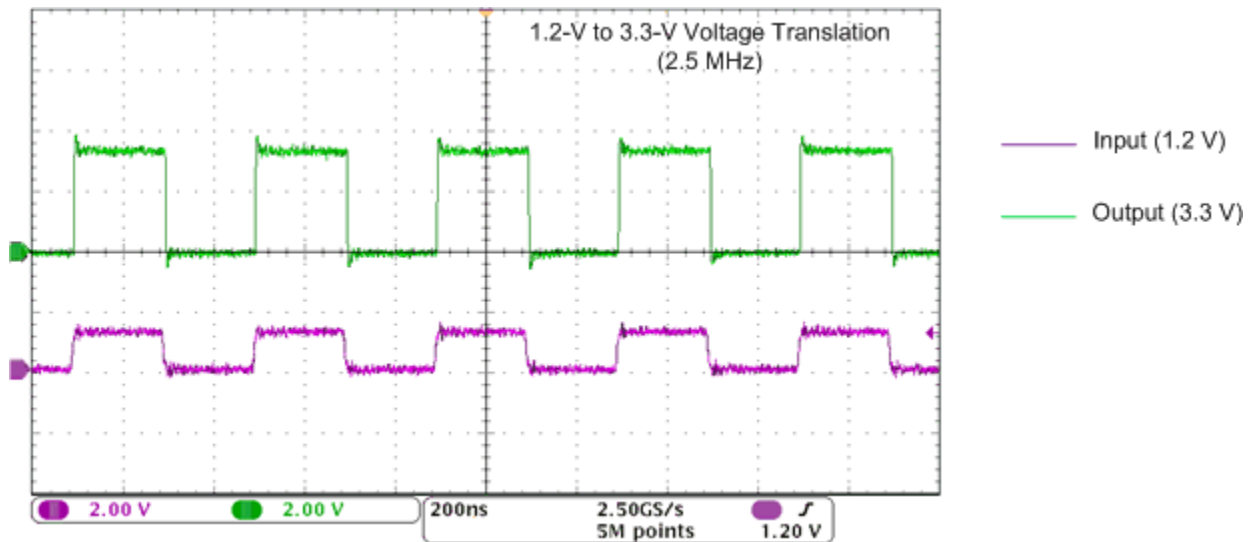


Figure 8-4. Translation Up (1.2V to 3.3V) at 2.5MHz

8.3 Power Supply Recommendations

The SN74AVC1T45 device uses two separate configurable power-supply rails, V_{CCA} and V_{CCB} . V_{CCA} accepts any supply voltage from 1.2V to 3.6V, and V_{CCB} accepts any supply voltage from 1.2V to 3.6V. The A port and B port are designed to track V_{CCA} and V_{CCB} respectively allowing for low-voltage, bidirectional translation between any of the 1.2V, 1.5 V, 1.8V, and 3.3V voltage nodes.

8.3.1 Power-Up Considerations

A proper power-up sequence always should be followed to avoid excessive supply current, bus contention, oscillations, or other anomalies. To guard against such power-up problems, take the following precautions:

1. Connect ground before any supply voltage is applied.
2. Power up V_{CCA} .
3. V_{CCB} can be ramped up along with or after V_{CCA} .

Table 8-3. Typical Total Static Power Consumption ($I_{CCA} + I_{CCB}$)

V_{CCB}	V_{CCA}						UNIT
	0V	1.2V	1.5V	1.8V	2.5V	3.3V	
0V	0	<0.5	<0.5	<0.5	<0.5	<0.5	μA
1.2V	<0.5	<1	<1	<1	<1	1	
1.5V	<0.5	<1	<1	<1	<1	1	
1.8V	<0.5	<1	<1	<1	<1	<1	
2.5V	<0.5	1	<1	<1	<1	<1	
3.3V	<0.5	1	<1	<1	<1	<1	

8.4 Layout

8.4.1 Layout Guidelines

For device reliability, it is recommended to follow common printed-circuit board layout guidelines such as follows:

- Bypass capacitors should be used on power supplies.
- Short trace lengths should be used to avoid excessive loading.
- Placing pads on the signal paths for loading capacitors or pullup resistors to help adjust rise and fall times of signals depending on the system requirements.

8.4.2 Layout Example

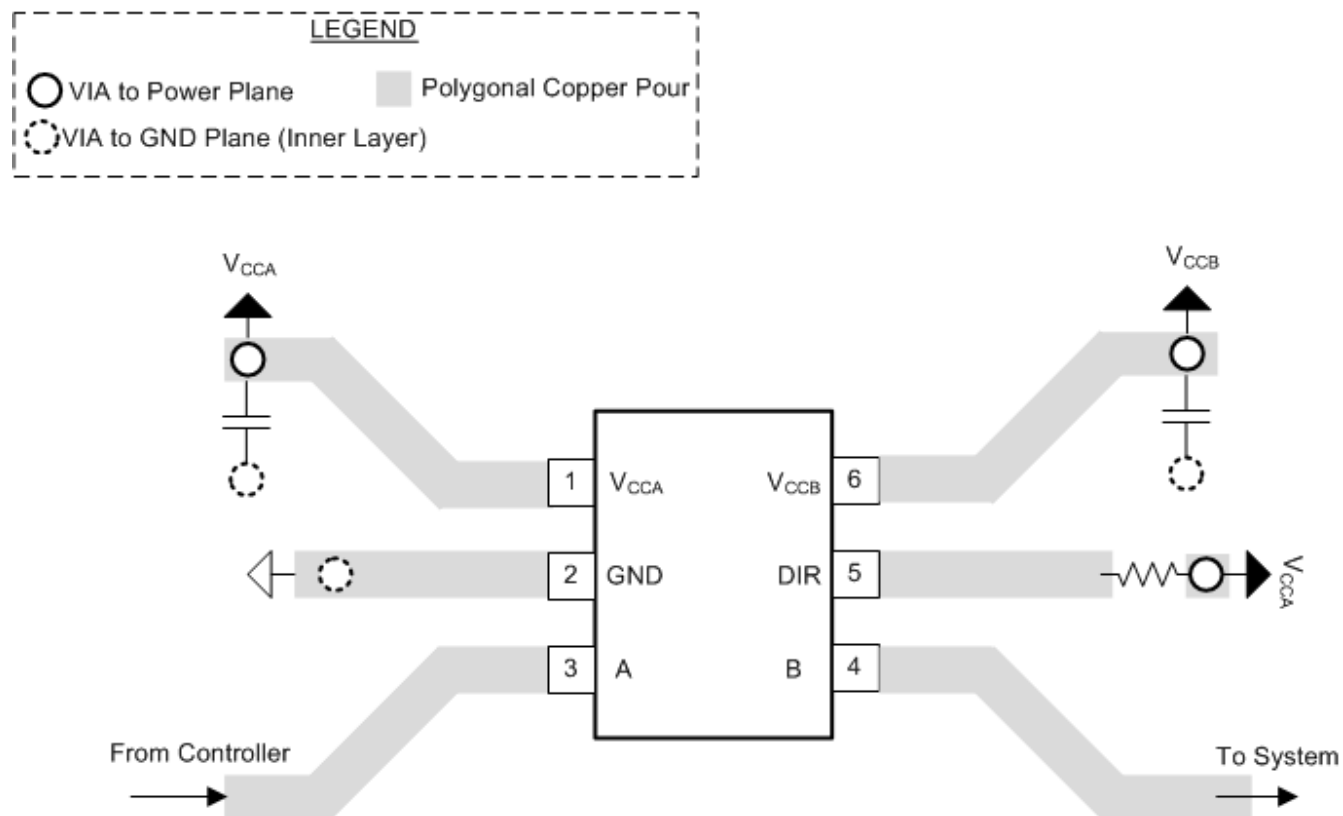


Figure 8-5. PCB Layout Example

9 Device and Documentation Support

9.1 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

9.2 Support Resources

[TI E2E™ support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

9.3 Trademarks

NanoFree™ is a trademark of Texas Instruments.

TI E2E™ is a trademark of Texas Instruments.

All trademarks are the property of their respective owners.

9.4 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

9.5 Glossary

[TI Glossary](#) This glossary lists and explains terms, acronyms, and definitions.

10 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision K (February 2025) to Revision L (January 2026)	Page
Updated 52ns to 5.2ns in t_{PHZ} row in the <i>Switching Characteristics</i>, $V_{CCA} = 1.2V$ table.....	9
Changes from Revision J (August 2024) to Revision K (February 2025)	Page
Updated DCK and DBV <i>Thermal Information</i>	8
Changes from Revision I (March 2024) to Revision J (August 2024)	Page
Updated Thermal Metrics.....	8
Changes from Revision H (October 2014) to Revision I (March 2024)	Page
- Updated the numbering format for tables, figures, and cross-references throughout the document..... - Updated the package information table to include package lead size.....	1 1
Changes from Revision G (January 2008) to Revision H (October 2014)	Page
Added <i>Pin Configuration and Functions</i> section, <i>ESD Ratings</i> table, <i>Feature Description</i> section, <i>Device Functional Modes</i>, <i>Application and Implementation</i> section, <i>Power Supply Recommendations</i> section, <i>Layout</i> section, <i>Device and Documentation Support</i> section, and <i>Mechanical, Packaging, and Orderable Information</i> section	1

11 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
SN74AVC1T45DBVR	Active	Production	SOT-23 (DBV) 6	3000 LARGE T&R	Yes	NIPDAU SN	Level-1-260C-UNLIM	-40 to 85	(DT1F, DT1R) (DT1H, DT1P)
SN74AVC1T45DBVR.A	Active	Production	SOT-23 (DBV) 6	3000 LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 85	(DT1F, DT1R) (DT1H, DT1P)
SN74AVC1T45DBVR.B	Active	Production	SOT-23 (DBV) 6	3000 LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 85	(DT1F, DT1R) (DT1H, DT1P)
SN74AVC1T45DBVRE4	Active	Production	SOT-23 (DBV) 6	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(DT1F, DT1R) (DT1H, DT1P)
SN74AVC1T45DBVRE4.A	Active	Production	SOT-23 (DBV) 6	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(DT1F, DT1R) (DT1H, DT1P)
SN74AVC1T45DBVRE4.B	Active	Production	SOT-23 (DBV) 6	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(DT1F, DT1R) (DT1H, DT1P)
SN74AVC1T45DBVRG4	Active	Production	SOT-23 (DBV) 6	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(DT1F, DT1R) (DT1H, DT1P)
SN74AVC1T45DBVRG4.A	Active	Production	SOT-23 (DBV) 6	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(DT1F, DT1R) (DT1H, DT1P)
SN74AVC1T45DBVRG4.B	Active	Production	SOT-23 (DBV) 6	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(DT1F, DT1R) (DT1H, DT1P)
SN74AVC1T45DBVT	Active	Production	SOT-23 (DBV) 6	250 SMALL T&R	Yes	Call TI Nipdau	Level-1-260C-UNLIM	-40 to 85	DT1R DT1H
SN74AVC1T45DBVT.B	Active	Production	SOT-23 (DBV) 6	250 SMALL T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	DT1R DT1H
SN74AVC1T45DBVTE4	Active	Production	SOT-23 (DBV) 6	250 SMALL T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	DT1R DT1H
SN74AVC1T45DCKR	Active	Production	SC70 (DCK) 6	3000 LARGE T&R	Yes	NIPDAU SN	Level-1-260C-UNLIM	-40 to 85	(TCF, TCR) (TCH, TCP)
SN74AVC1T45DCKR.A	Active	Production	SC70 (DCK) 6	3000 LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 85	(TCF, TCR) (TCH, TCP)
SN74AVC1T45DCKR.B	Active	Production	SC70 (DCK) 6	3000 LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 85	(TCF, TCR) (TCH, TCP)
SN74AVC1T45DCKRE4	Active	Production	SC70 (DCK) 6	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(TCF, TCR) (TCH, TCP)

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
SN74AVC1T45DCKRE4.A	Active	Production	SC70 (DCK) 6	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(TCF, TCR) (TCH, TCP)
SN74AVC1T45DCKRE4.B	Active	Production	SC70 (DCK) 6	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(TCF, TCR) (TCH, TCP)
SN74AVC1T45DCKRG4	Active	Production	SC70 (DCK) 6	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(TCF, TCR) (TCH, TCP)
SN74AVC1T45DCKRG4.A	Active	Production	SC70 (DCK) 6	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(TCF, TCR) (TCH, TCP)
SN74AVC1T45DCKRG4.B	Active	Production	SC70 (DCK) 6	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(TCF, TCR) (TCH, TCP)
SN74AVC1T45DCKT	Active	Production	SC70 (DCK) 6	250 SMALL T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(TCF, TCR) (TCH, TCP)
SN74AVC1T45DCKT.B	Active	Production	SC70 (DCK) 6	250 SMALL T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(TCF, TCR) (TCH, TCP)
SN74AVC1T45DCKTG4	Active	Production	SC70 (DCK) 6	250 SMALL T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(TCF, TCR) (TCH, TCP)
SN74AVC1T45DRLR	Active	Production	SOT-5X3 (DRL) 6	4000 LARGE T&R	Yes	NIPDAU NIPDAUAG	Level-1-260C-UNLIM	-40 to 85	(1JW, TCR) TCH
SN74AVC1T45DRLR.A	Active	Production	SOT-5X3 (DRL) 6	4000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(1JW, TCR) TCH
SN74AVC1T45DRLR.B	Active	Production	SOT-5X3 (DRL) 6	4000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(1JW, TCR) TCH
SN74AVC1T45DRLRG4	Active	Production	SOT-5X3 (DRL) 6	4000 LARGE T&R	Yes	NIPDAU NIPDAUAG	Level-1-260C-UNLIM	-40 to 85	(1JW, TCR) TCH
SN74AVC1T45DRLRG4.A	Active	Production	SOT-5X3 (DRL) 6	4000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(1JW, TCR) TCH
SN74AVC1T45DRLRG4.B	Active	Production	SOT-5X3 (DRL) 6	4000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	(1JW, TCR) TCH
SN74AVC1T45YZPR	Active	Production	DSBGA (YZP) 6	3000 LARGE T&R	Yes	SNAGCU	Level-1-260C-UNLIM	-40 to 85	(TC2, TCN)
SN74AVC1T45YZPR.B	Active	Production	DSBGA (YZP) 6	3000 LARGE T&R	Yes	SNAGCU	Level-1-260C-UNLIM	-40 to 85	(TC2, TCN)

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

(2) Material type: When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) RoHS values: Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

(4) Lead finish/Ball material: Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) MSL rating/Peak reflow: The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) Part marking: There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

Important Information and Disclaimer: The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

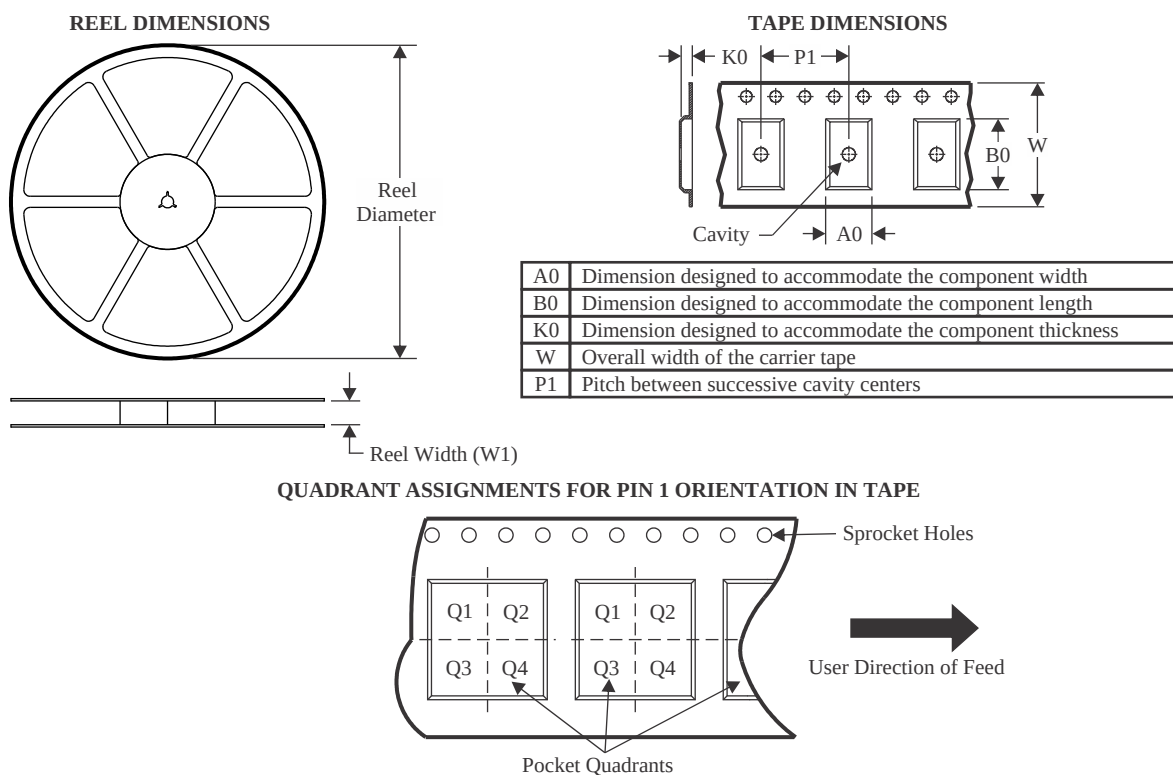
OTHER QUALIFIED VERSIONS OF SN74AVC1T45 :

- Automotive : [SN74AVC1T45-Q1](#)

NOTE: Qualified Version Definitions:

- Automotive - Q100 devices qualified for high-reliability automotive applications targeting zero defects

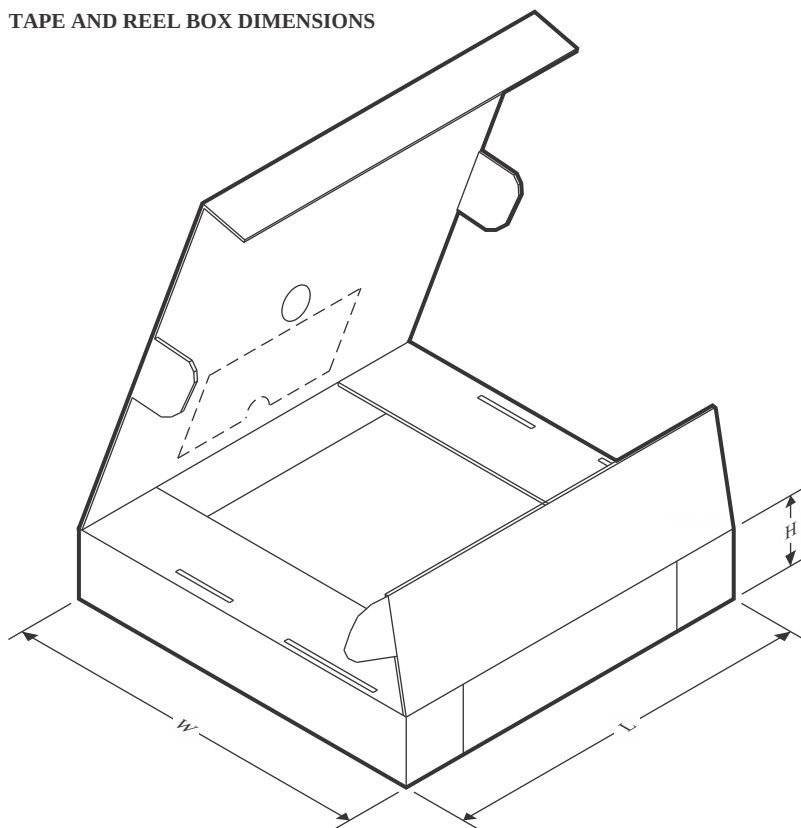
TAPE AND REEL INFORMATION



*All dimensions are nominal

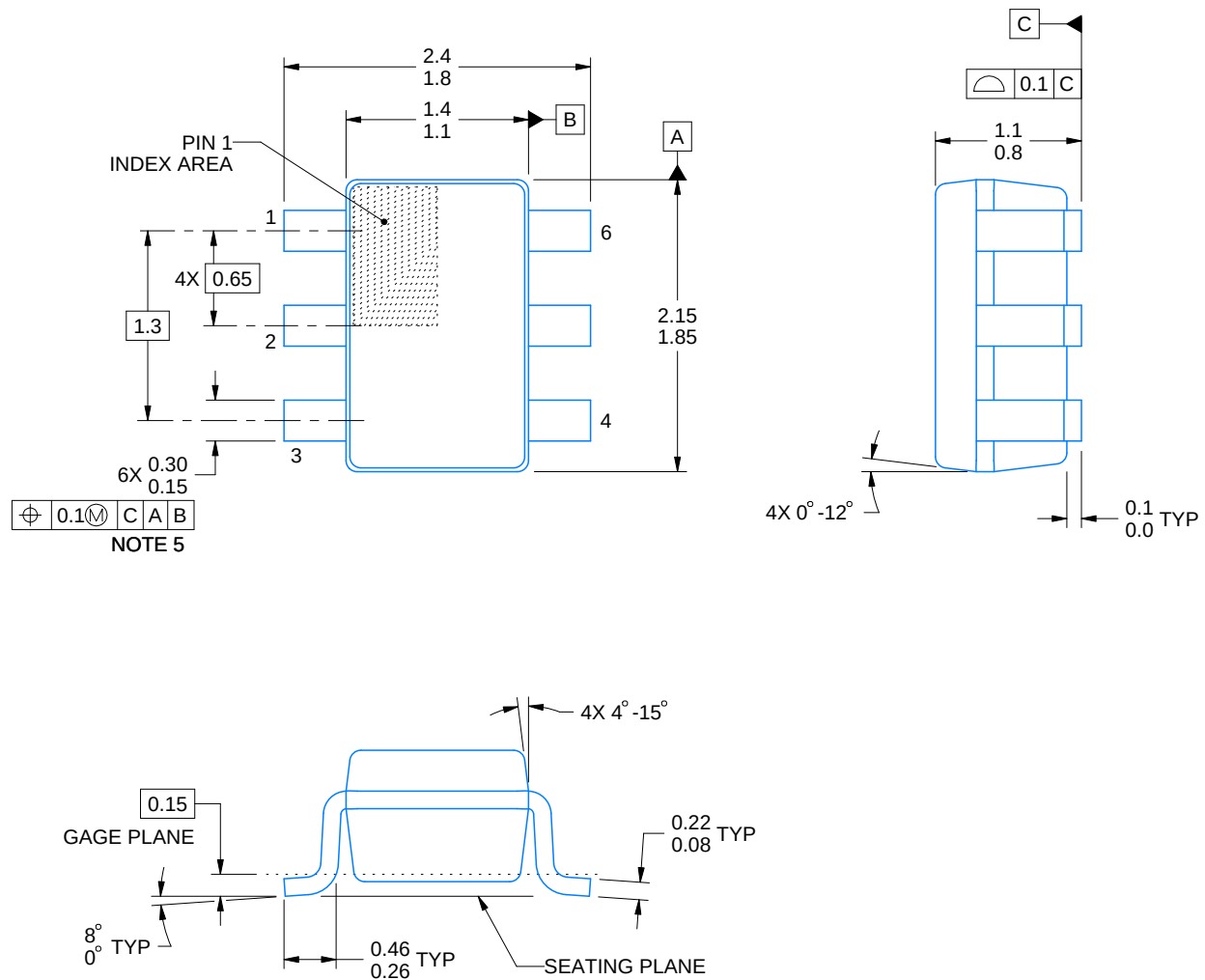
Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
SN74AVC1T45DBVR	SOT-23	DBV	6	3000	180.0	8.4	3.23	3.17	1.37	4.0	8.0	Q3
SN74AVC1T45DBVR	SOT-23	DBV	6	3000	178.0	8.4	3.2	3.2	1.4	4.0	8.0	Q3
SN74AVC1T45DBVR	SOT-23	DBV	6	3000	180.0	8.4	3.2	3.2	1.4	4.0	8.0	Q3
SN74AVC1T45DBVRE4	SOT-23	DBV	6	3000	180.0	8.4	3.23	3.17	1.37	4.0	8.0	Q3
SN74AVC1T45DBVRG4	SOT-23	DBV	6	3000	180.0	8.4	3.23	3.17	1.37	4.0	8.0	Q3
SN74AVC1T45DBVT	SOT-23	DBV	6	250	180.0	8.4	3.23	3.17	1.37	4.0	8.0	Q3
SN74AVC1T45DCKR	SC70	DCK	6	3000	180.0	8.4	2.41	2.41	1.2	4.0	8.0	Q3
SN74AVC1T45DCKR	SC70	DCK	6	3000	178.0	8.4	2.25	2.45	1.2	4.0	8.0	Q3
SN74AVC1T45DCKRE4	SC70	DCK	6	3000	180.0	8.4	2.41	2.41	1.2	4.0	8.0	Q3
SN74AVC1T45DCKRG4	SC70	DCK	6	3000	180.0	8.4	2.41	2.41	1.2	4.0	8.0	Q3
SN74AVC1T45DCKT	SC70	DCK	6	250	180.0	8.4	2.41	2.41	1.2	4.0	8.0	Q3
SN74AVC1T45DRLR	SOT-5X3	DRL	6	4000	180.0	8.4	1.98	1.78	0.69	4.0	8.0	Q3
SN74AVC1T45DRLR	SOT-5X3	DRL	6	4000	180.0	8.4	2.0	1.8	0.75	4.0	8.0	Q3
SN74AVC1T45DRLRG4	SOT-5X3	DRL	6	4000	180.0	8.4	2.0	1.8	0.75	4.0	8.0	Q3
SN74AVC1T45YZPR	DSBGA	YZP	6	3000	178.0	9.2	1.02	1.52	0.63	4.0	8.0	Q1

TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

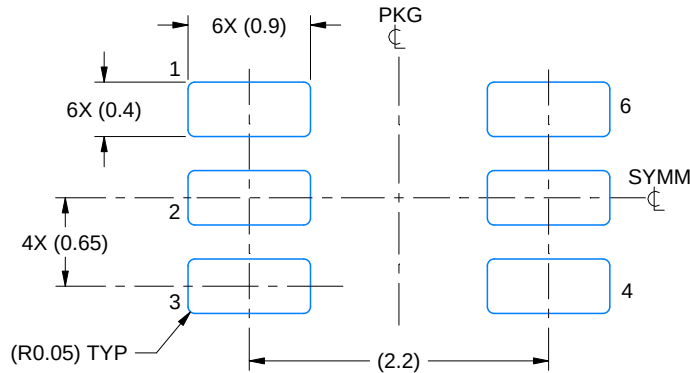
Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
SN74AVC1T45DBVR	SOT-23	DBV	6	3000	202.0	201.0	28.0
SN74AVC1T45DBVR	SOT-23	DBV	6	3000	208.0	191.0	35.0
SN74AVC1T45DBVR	SOT-23	DBV	6	3000	210.0	185.0	35.0
SN74AVC1T45DBVRE4	SOT-23	DBV	6	3000	202.0	201.0	28.0
SN74AVC1T45DBVRG4	SOT-23	DBV	6	3000	202.0	201.0	28.0
SN74AVC1T45DBVT	SOT-23	DBV	6	250	202.0	201.0	28.0
SN74AVC1T45DCKR	SC70	DCK	6	3000	202.0	201.0	28.0
SN74AVC1T45DCKR	SC70	DCK	6	3000	208.0	191.0	35.0
SN74AVC1T45DCKRE4	SC70	DCK	6	3000	202.0	201.0	28.0
SN74AVC1T45DCKRG4	SC70	DCK	6	3000	202.0	201.0	28.0
SN74AVC1T45DCKT	SC70	DCK	6	250	202.0	201.0	28.0
SN74AVC1T45DRLR	SOT-5X3	DRL	6	4000	202.0	201.0	28.0
SN74AVC1T45DRLR	SOT-5X3	DRL	6	4000	210.0	185.0	35.0
SN74AVC1T45DRLRG4	SOT-5X3	DRL	6	4000	210.0	185.0	35.0
SN74AVC1T45YZPR	DSBGA	YZP	6	3000	220.0	220.0	35.0



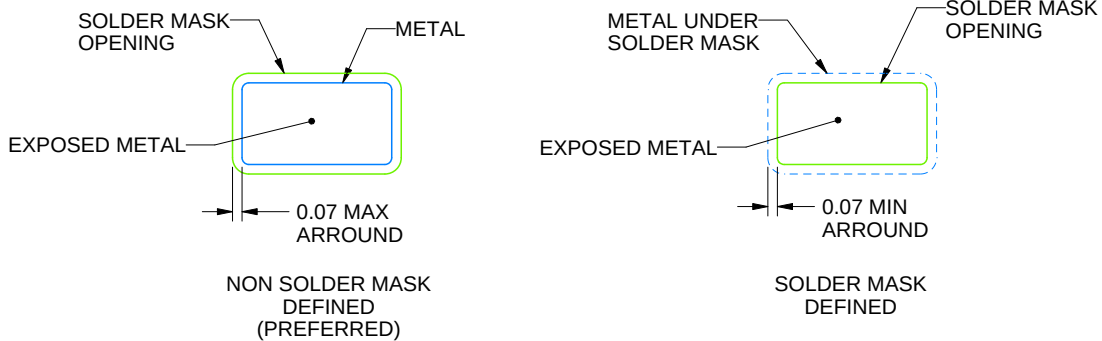
4214835/D 11/2024

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Body dimensions do not include mold flash or protrusion. Mold flash and protrusion shall not exceed 0.15 per side.
4. Falls within JEDEC MO-203 variation AB.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:18X

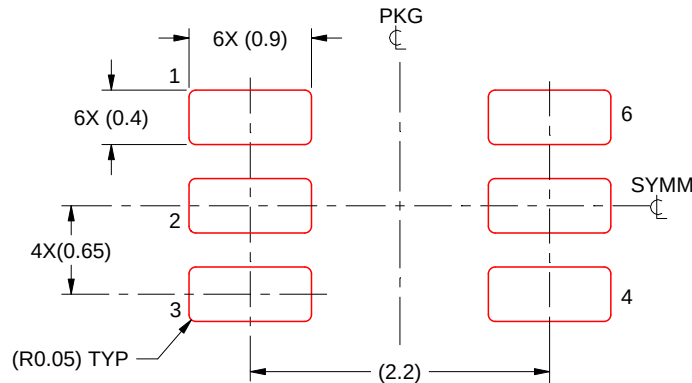


SOLDER MASK DETAILS

4214835/D 11/2024

NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
BASED ON 0.125 THICK STENCIL
SCALE:18X

4214835/D 11/2024

NOTES: (continued)

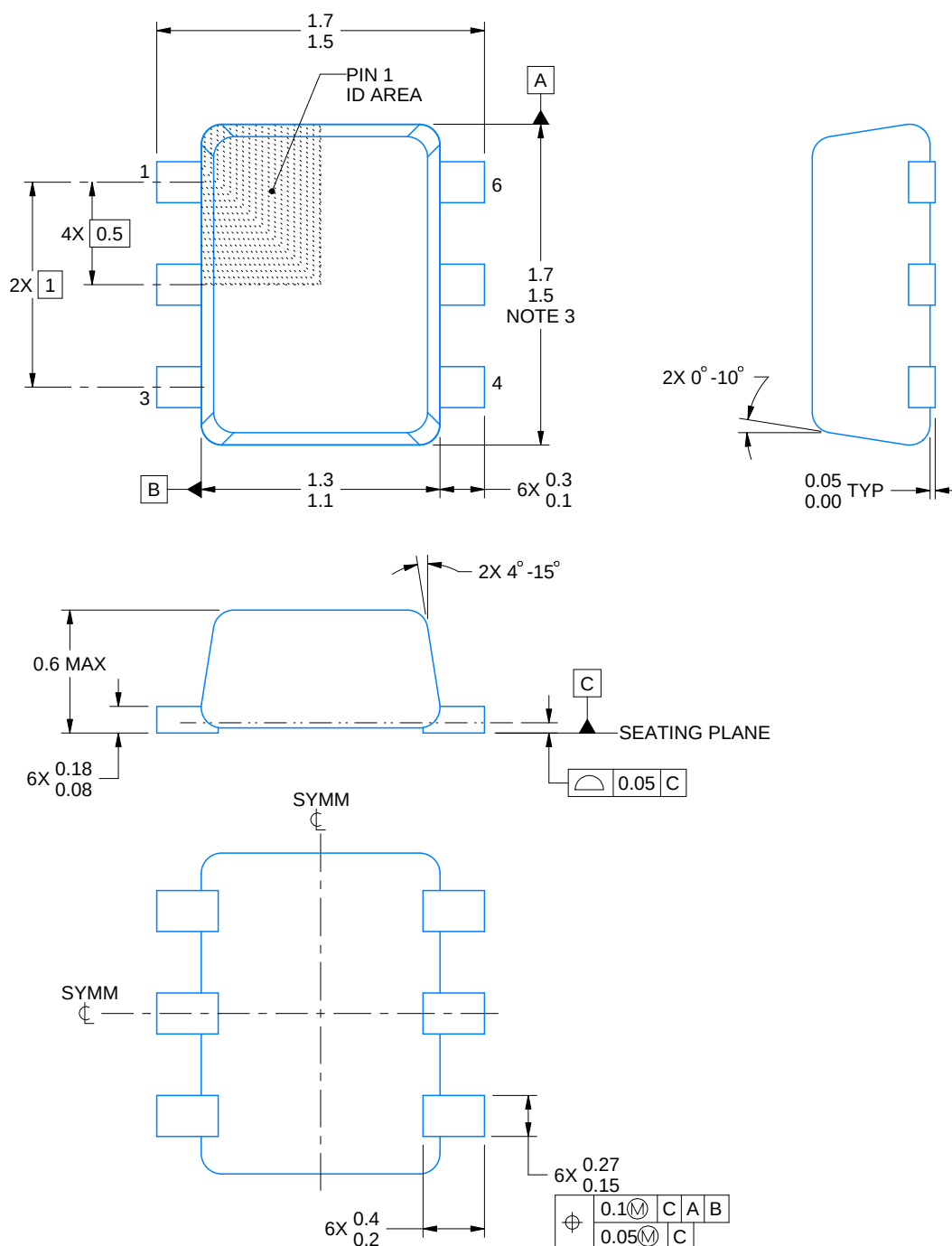
7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.



PACKAGE OUTLINE

SOT - 0.6 mm max height

PLASTIC SMALL OUTLINE



4223266/F 11/2024

NOTES:

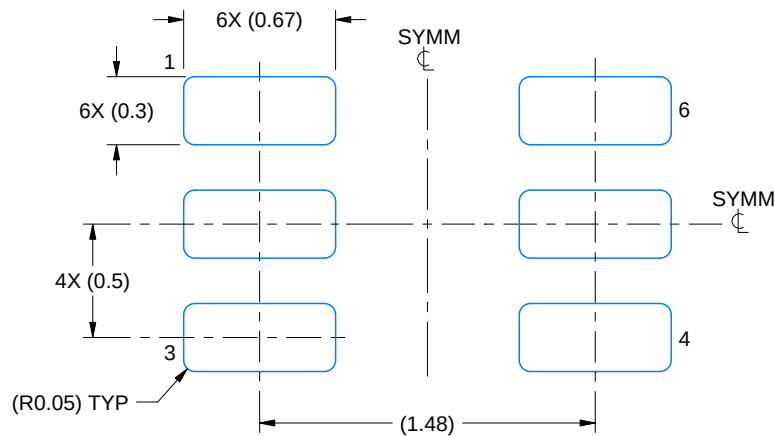
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. Reference JEDEC registration MO-293 Variation UAAD

EXAMPLE BOARD LAYOUT

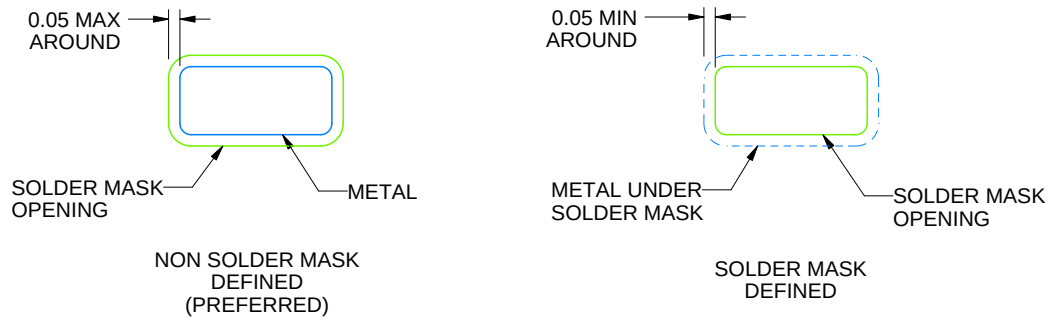
DRL0006A

SOT - 0.6 mm max height

PLASTIC SMALL OUTLINE



LAND PATTERN EXAMPLE
SCALE:30X



SOLDERMASK DETAILS

4223266/F 11/2024

NOTES: (continued)

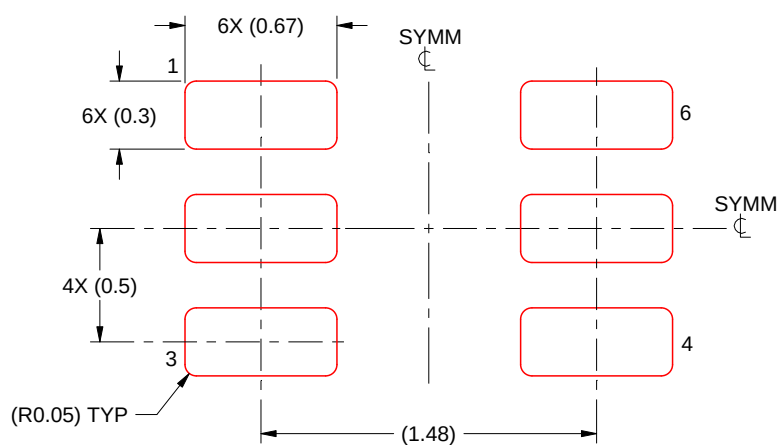
5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.
7. Land pattern design aligns to IPC-610, Bottom Termination Component (BTC) solder joint inspection criteria.

EXAMPLE STENCIL DESIGN

DRL0006A

SOT - 0.6 mm max height

PLASTIC SMALL OUTLINE



SOLDER PASTE EXAMPLE
BASED ON 0.1 mm THICK STENCIL
SCALE:30X

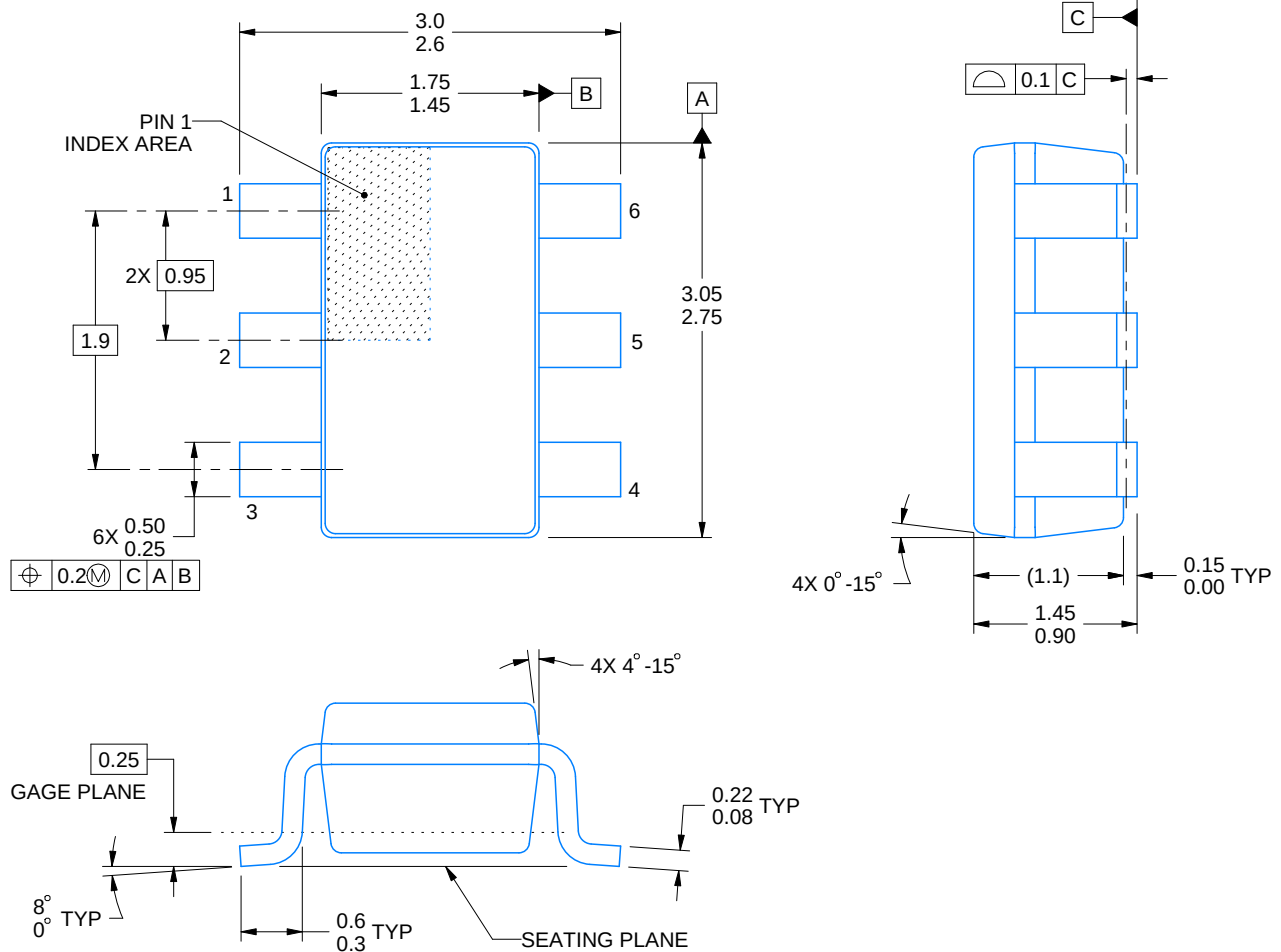
4223266/F 11/2024

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

DBV0006A**PACKAGE OUTLINE****SOT-23 - 1.45 mm max height**

SMALL OUTLINE TRANSISTOR



4214840/G 08/2024

NOTES:

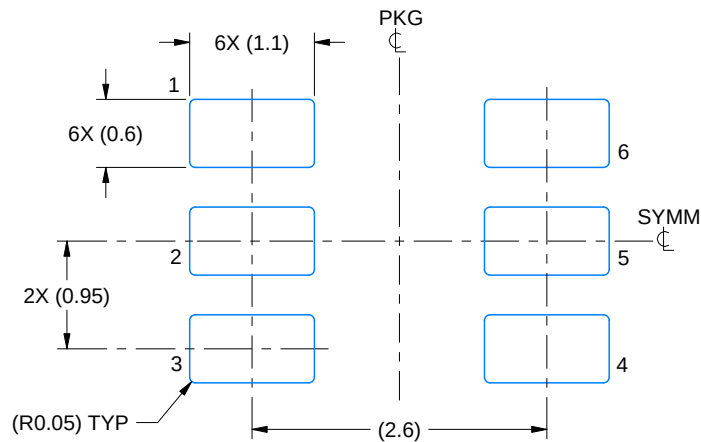
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Body dimensions do not include mold flash or protrusion. Mold flash and protrusion shall not exceed 0.25 per side.
4. Leads 1,2,3 may be wider than leads 4,5,6 for package orientation.
5. Reference JEDEC MO-178.

EXAMPLE BOARD LAYOUT

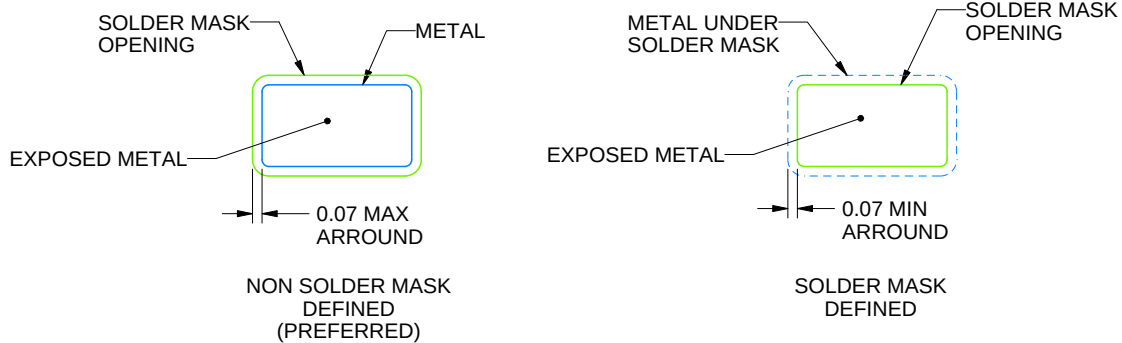
DBV0006A

SOT-23 - 1.45 mm max height

SMALL OUTLINE TRANSISTOR



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:15X



SOLDER MASK DETAILS

4214840/G 08/2024

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

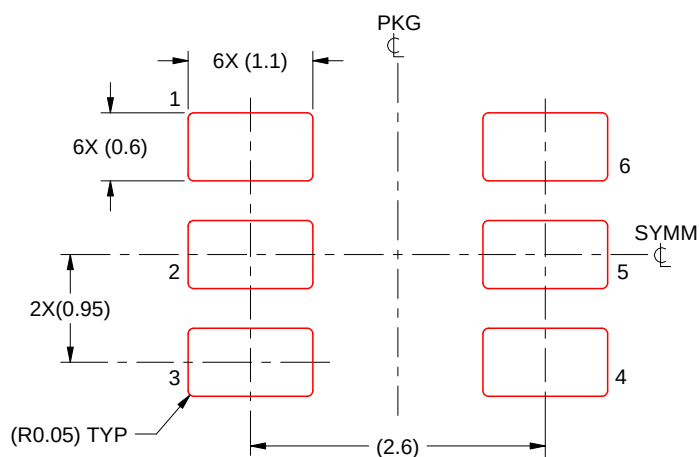
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

DBV0006A

SOT-23 - 1.45 mm max height

SMALL OUTLINE TRANSISTOR



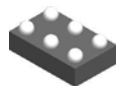
SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:15X

4214840/G 08/2024

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

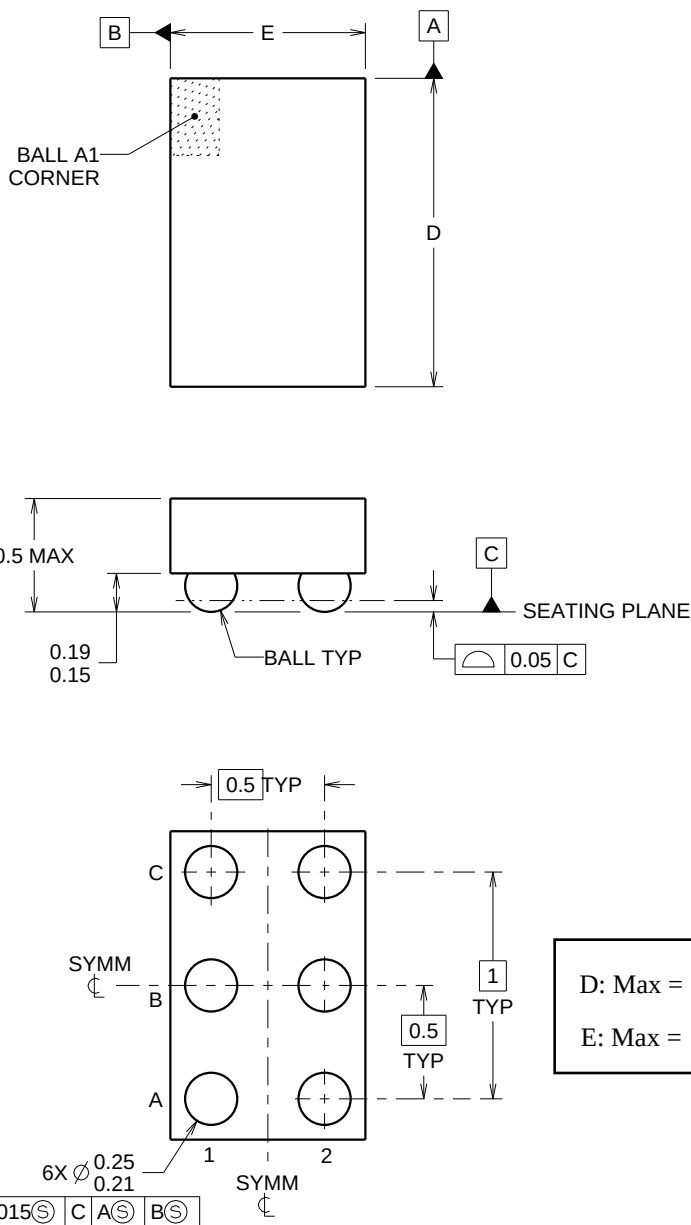
YZP0006



PACKAGE OUTLINE

DSBGA - 0.5 mm max height

DIE SIZE BALL GRID ARRAY



4219524/A 06/2014

NOTES:

NanoFree Is a trademark of Texas Instruments.

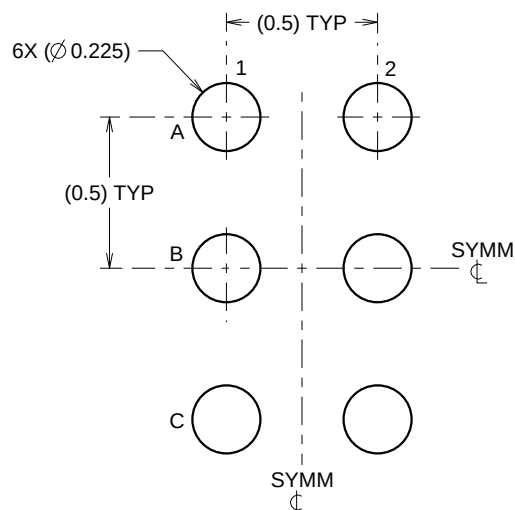
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. NanoFree™ package configuration.

EXAMPLE BOARD LAYOUT

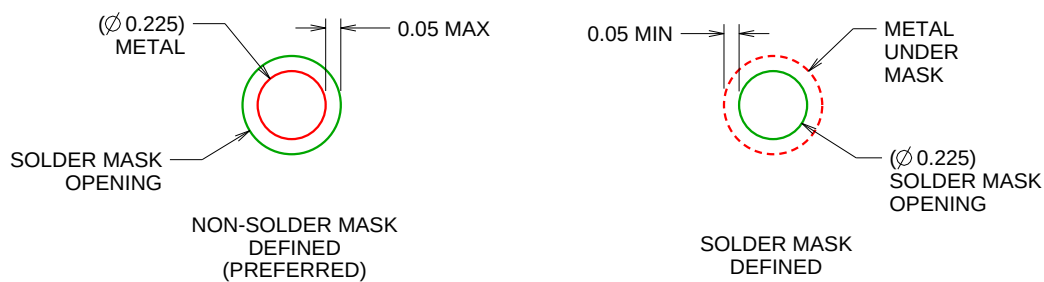
YZP0006

DSBGA - 0.5 mm max height

DIE SIZE BALL GRID ARRAY



LAND PATTERN EXAMPLE
SCALE:40X



SOLDER MASK DETAILS
NOT TO SCALE

4219524/A 06/2014

NOTES: (continued)

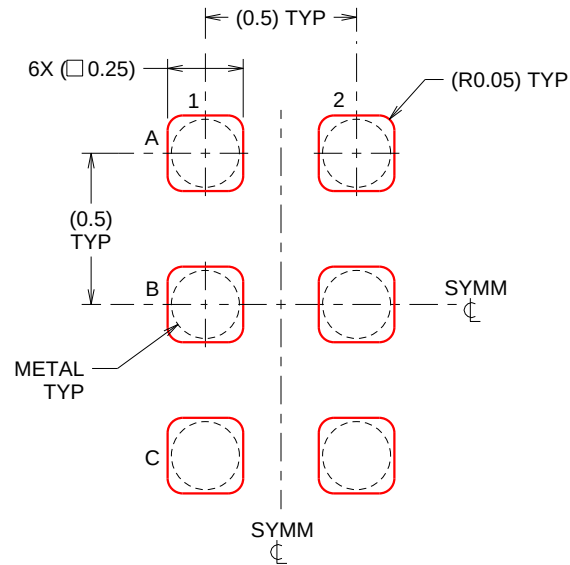
4. Final dimensions may vary due to manufacturing tolerance considerations and also routing constraints.
For more information, see Texas Instruments literature number SBVA017 (www.ti.com/lit/sbva017).

EXAMPLE STENCIL DESIGN

YZP0006

DSBGA - 0.5 mm max height

DIE SIZE BALL GRID ARRAY



SOLDER PASTE EXAMPLE
BASED ON 0.1 mm THICK STENCIL
SCALE:40X

4219524/A 06/2014

NOTES: (continued)

5. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release.

IMPORTANT NOTICE AND DISCLAIMER

TI PROVIDES TECHNICAL AND RELIABILITY DATA (INCLUDING DATASHEETS), DESIGN RESOURCES (INCLUDING REFERENCE DESIGNS), APPLICATION OR OTHER DESIGN ADVICE, WEB TOOLS, SAFETY INFORMATION, AND OTHER RESOURCES "AS IS" AND WITH ALL FAULTS, AND DISCLAIMS ALL WARRANTIES, EXPRESS AND IMPLIED, INCLUDING WITHOUT LIMITATION ANY IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR NON-INFRINGEMENT OF THIRD PARTY INTELLECTUAL PROPERTY RIGHTS.

These resources are intended for skilled developers designing with TI products. You are solely responsible for (1) selecting the appropriate TI products for your application, (2) designing, validating and testing your application, and (3) ensuring your application meets applicable standards, and any other safety, security, regulatory or other requirements.

These resources are subject to change without notice. TI grants you permission to use these resources only for development of an application that uses the TI products described in the resource. Other reproduction and display of these resources is prohibited. No license is granted to any other TI intellectual property right or to any third party intellectual property right. TI disclaims responsibility for, and you fully indemnify TI and its representatives against any claims, damages, costs, losses, and liabilities arising out of your use of these resources.

TI's products are provided subject to [TI's Terms of Sale](#), [TI's General Quality Guidelines](#), or other applicable terms available either on ti.com or provided in conjunction with such TI products. TI's provision of these resources does not expand or otherwise alter TI's applicable warranties or warranty disclaimers for TI products. Unless TI explicitly designates a product as custom or customer-specified, TI products are standard, catalog, general purpose devices.

TI objects to and rejects any additional or different terms you may propose.

Copyright © 2026, Texas Instruments Incorporated

Last updated 10/2025